Israel Souza Ferreira

Data Scientist | Machine Learning Engineer

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PROFILE

Data scientist and machine learning engineer experienced in **applied research** and end-to-end delivery: experimentation, computer vision, APIs, data, and CPU-only deployment. I turn hypotheses into measurable decisions through baselines, ablations, benchmarks, and automated testing.

PROFESSIONAL EXPERIENCE

Tieta Artificial Intelligence — **Remote, Brazil**
R&D Researcher — Research and Innovation Training Program — *Jul 2026 - Present*

- Developing an **OCR and document-segmentation pipeline** that turns scanned exams into structured questions for educational analysis.

Tieta Artificial Intelligence — **Remote, Brazil**
Data Scientist / R&D Researcher — CNPq RHAE Research Fellow — *Apr 2025 - Jun 2026*

- Modeled psychometric parameters with **Item Response Theory** and compared approaches through paired evaluations, ablations, correlation with references, and inference cost.
- Built signal-extraction pipelines with LLMs orchestrated by **DSPy** and evaluated multitask regression architectures over textual and visual data through reproducible experiments.

Independent Consulting — Educational Institution — **Remote, Brazil**
Full Stack Developer and Machine Learning Engineer — *2026 - Present*

- Built the **Face Clock Evoluir** end to end, a contracted facial attendance system in final acceptance testing, covering models, backend, database, security, testing, and deployment.
- Put 1:N recognition into CPU-only operation with YuNet + SFace, ONNX Runtime, and PostgreSQL/pgvector: **74 ms p95** with one identification and **574 automated tests** in the current delivery.
- Recalibrated the open-set threshold, reducing accepted unknowns from **45.4% to 0.2%** on public reference data; reduced **5 to 3 frames** after identical accuracy from 1 to 8, and fixed a race causing **24/24 failures** across eight simultaneous bursts.
- A vector-search benchmark measured **0.076 ms over 1,000 vectors**; because HNSW reached only 0.915 recall@1, I kept exact, transactional pgvector search instead of adopting FAISS prematurely.
- Compared five approaches for medical-document triage; EfficientNet + Logistic Regression reached **87.5% accuracy** with seven fewer false positives at the same recall.
- Fit the delivery to an 8 GB, GPU-less server: production images fell from **10.59 to 1.11 GB** (9.5× smaller), with zero classifier decisions changed in the migration to ONNX Runtime.

Federal University of Ceará — Quixadá Campus — **On-site, Brazil**
Teaching Assistant for Algorithms, Calculus, and Pre-Calculus — *2022 - 2024*

- Supported classes in data structures, algorithms, complexity, and Calculus, translating quantitative material across skill levels.

SELECTED PROJECTS

- **Triple Roman Domination in Graphs (undergraduate thesis):** first GA- and ACO-based metaheuristics reported for the problem, plus a correction to the published ILP formulation; evaluated over **362 graphs**.
- **Brazilian Emergency Aid — Data Engineering:** chunked binary `COPY` ingestion over **31.6 GB** of public data; instrumentation linked memory growth to retained deduplication IDs (**r = 0.99**).

TECHNICAL SKILLS

Languages and querying: Python, SQL, C++, and Bash.

ML, vision, and experimentation: PyTorch, Scikit-learn, OpenCV, YuNet, SFace, ONNX Runtime, DSPy, Optuna, cross-validation, calibration, ablation, hypothesis testing, and vector search.

Data and production: PostgreSQL, pgvector, FAISS, Pandas, NumPy, SciPy, FastAPI, SQLAlchemy, Alembic, Docker, GitHub Actions, pytest, MLflow, and Streamlit.

EDUCATION

Federal University of Ceará — Quixadá Campus — **Brazil**
B.Sc. in Computer Science — *Jul 2026*

LANGUAGES

Portuguese: Native | **English:** Fluent technical reading of research papers and documentation